Preparation of culture media

What is Culture Media?

The media is a source of nutrients to support the growth of the micro-organisms *in vitro*. The media helps in the growth and counting of microbial cells, selection of microorganisms, and survival of microorganisms. The culture medium can be liquid or gel.

Components of culture media

1. **Carbon sources** may be either monosaccharides (such as glucose and fructose), or they may be disaccharides (sucrose and maltose), polysaccharides such as starch.

2**. Nitrogen sources**, either from organic sources such as peptone and amino acids, or from inorganic sources such as nitrate salts

3**. Salts** have a significant role in regulating the osmotic pressure inside the cell. Either it is major such as the ammonium salts Na, Ca, P, K, or it is a minor micronutrient such as. Zn, Fe,Mn ,Bo, Cu

4 **. Humidity** Acceptable is 10%

5. **Vitamins**, most fungi produce them.

The importance of using microbial media:

1. Development and conservation of species for microorganisms.
2. Induction of microorganisms to the production or composition of some substances.
3. To culture microbes.
4. To identify the cause of infection.
5. To identify characteristics of microorganisms.
6. To isolate pure culture.
7. To store the culture stock.
8. To observe biochemical reactions.
9. To test microbial contamination in any sample.

10. To check antimicrobial agents and preservatives effect.
11. To observe microbe colony type, its color, shape, cause.
12. To differentiate between different colonies.
13. To create antigens for laboratory use.
14. To estimate viable count.
15. To test antibiotic sensitivity.

Types of culture media based on chemical composition/application:

There are seven routine laboratory media.

1. Basal media
2. Enriched media
3. Selective media
4. Enrichment media
5. Indicator media or differential media.
6. Transport media
7. Storage media.

How to prepare culture media:

1. Weigh the amount of ingredients powder on weighing balance.
2. Dissolve the ingredients in distilled water.
3. Adjust pH of the medium if needed.
4. Add agar and boiled or cooking it to dissolve.
5. Pour the media into flask.
6. Autoclave the media when ingredients fully dissolve. Sterilization is done in autoclave to prevent from contamination, at 121ºC for 15-20 min at 15lbs.

7. After the autoclave place the media flask in laminar air flow to cool (45-50 ºC).
8. Sterilize the laminar air flow with 70% alcohol.
9. A bit cools down the media and pours into sterile Petri-plates for solidification.
10. Then sample is ready to spread by (spreader) / streak by (Inoculation loop) on the medium for identification or isolation of microbes.
11. Sealed the Petri plates with paraffin, label them.
12. Keep them inverted in incubator at 37ºC for 24hrs.
13. Observe the result next day colonies formation is visible on the media.

Isolation of bacteria

There are two main methods of direct plate counting: spread plate; and pour plate method:

1-The spread plate method consists of evenly spreading the diluted sample over an agar plate. Using this method yields colonies that form on the surface of the agar.

procedure:

1. make serial dilution of microorganism sample in series of tubes containing D.W.
2. transfer 0.1ml from last dilution of microorganism culture by pipette.
3. put it on the center of an agar plate.
4. moist spreader with alcohol and sterilize by flaming .
5. spread the sample on agar plat by spreader.
6. sterilize it again.
7. incubate the plate at 37°c for 24 hours, and then examine and count the present colonies distributed throughout the agar.

2-The pour plate method:

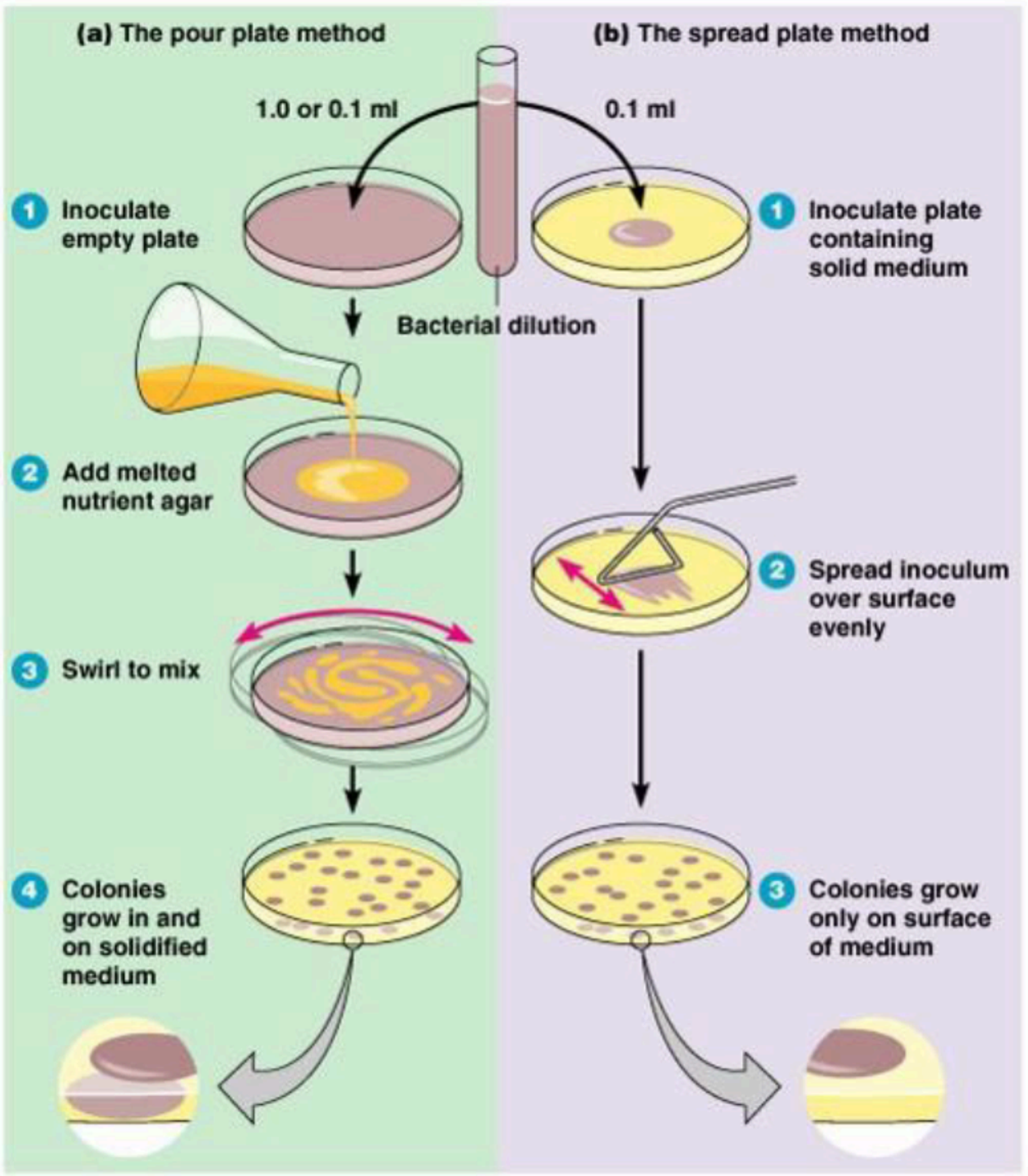
A volume of 1 ml of the diluted sample is put into a sterile petri plate, then melted agar is poured in and mixed with the sample. This method yields colonies that form colonies throughout the agar (growing both on the agar and in the agar , not just on the surface.

Procedure:

1. Put agar media in water bath in 45°c. to liquefied.
2. Add 1gm of sample to first tube and make serial dilution from one to another tube.
3. transfer 1ml from last dilution of microorganism culture by pipette ,then Put in sterile petri dish.
4. Pour melted agar and mixed with the dilution sample.
5. Leave petri dish to solidify .
6. Incubate the plate at 37°c for 24 hour Determine No. of bacterial cells in soil sample from equation:

No. of bacterial cells /1gm moist soil =No. of colonies × inverted dilution .

No. of bacterial cells /1gm dry soil = No. of colonies ×inverted dilution



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3. Streak Plate Method:

The streak plate method is based on the principle of dilution. It can be described as a rapid qualitative isolation technique. The main criterion of isolation is to obtain a reduced number of colonies. In this technique, a loopful of culture is spread on an agar plate to get individual cells far apart enough from each other. The streaking method gradually dilutes the inoculum such that the bacterial cells can be counted as colony forming units (CFUs).

